Ammar Akhlaq Ahmed

Ammar.akhlaqahmed1@gmail.com|GitHub|Linktree|

MSc Systems & Control student and mathematician specializing in nonlinear, robust, and Port-Hamiltonian control for robotics and energy systems. Experienced in developing a ROS 2-based sim-to-real Unitree G1 grasping stack, system-identification tools, and MATLAB/Simulink control applications. Skilled in Python, ROS 2, Isaac Sim, MuJoCo, URDF, kinematics, and Git. Beyond academics, I find inspiration in talking to people and discovering new paradigms and enjoying travelling.

EDUCATION

University Of Groningen ***Netherlands*** | *September 2024 – Current*
MASTERS APPLIED MATHEMATICS SYSTEMS & CONTROL
Nonlinear control, PLC Design, Robust control, Model reductions, Elementary Plasma Physics, Data Identification, Model Identification & Digital twin theory, Humanoid Robotics

FORMAN CHRISTIAN COLLEGE SCHOOL OF MATHEMATICS ***Lahore Pakistan*** | *September 2018 -April 2023*
Bachelor of Science (BSc) Mathematics
Courses: Partial Differential Equations, Topology, and Metric spaces, Electricity and Magnetism, Set Theory, Mechanics, Wave, and Vibrations, Vector Analysis Group Analysis.

WORK EXPERIENCE & PROJECTS

Thesis Project, *Passivity Based Humanoid Grasping* **Netherlands** |*04/2025-present*

- Designed a white-box manipulation stack for the Unitree G1 humanoid robot to perform grasping, pick-up, transport, and placement tasks.
- Developed URDF-driven automatic generation of Denavit–Hartenberg parameters, forward kinematics, geometric 6D Jacobians, and damped least-squares inverse kinematics for robust end-effector control. Formulated the hand and arm dynamics as a Port-Hamiltonian control system using the Unitree Python SDK.
- Implemented impedance control for the hand and PID-based arm control with gravity compensation, force-regulated grasping, and energy-based stability interpretation. Built a complete ROS 2 pipeline with MuJoCo and Isaac Sim simulation backends, integrating camera-based object localisation, trajectory orchestration, RViz2 visualisation, TF2 transformations, and closed-loop torque control.
- Transferred grasping and pick-and-place behaviors from simulation to Unitree G1 hardware and investigated robustness under model mismatch.
- Performed Ishikawa Analysis of ROS 2 integration issues involving simulation to real application, TF trees, URDF inconsistencies, contact modelling, and controller interfaces.
- Developed reusable system-integration and troubleshooting documentation using Markdown, Obsidian, GitHub, and version-controlled project workflows. Structured the manipulation stack into perception, kinematics, planning, simulation, and control subsystems to support reproducibility and future verification activities.

EVEREST BIOCHEM, Junior *Supply Chain Manager-Business Developer* ***Lahore, Pakistan*** |*23/03/2023 – 01/07/2024*

- Managed procurement, supplier coordination, commissioning, and operation of a 2-ton ribbon blender used in industrial production and development of a sizing-agent product line generating approximately PKR 5 million in monthly revenue.
- Developed and evaluated mechanical concept designs for twin-screw and ribbon-blender equipment using Blender.
- Supported design validation using ANSYS before equipment commissioning and manufacturing decisions.
- Coordinated equipment commissioning, operational readiness, safety activities, and material handling procedures.
- Developed safe-management procedures for material handling, packaging, and storage using plastic drums and other industrial packaging systems.
- Prepared comprehensive procurement, commissioning, operating, safety, and process documents using Microsoft Office and SAP.
- Conducted market research for the textile industry to identify customer needs, production requirements, and machinery opportunities.
- Evaluated textile-manufacturing equipment requirements and supported business-development decisions for new product and process initiatives.
- Developed a requirements-engineering workflow for engineering projects to improve requirement definition, traceability, system verification, and implementation of quality.
- Established connections with manufacturing companies and coordinated stakeholders to support the development of logistics and production systems.
- Established Lab procedure for product-testing function to evaluate new products and process materials using twin-screw and extrusion machinery.
- Developed test-management activities covering test planning, test-method development, test execution, test records, results review, and technical reporting.
- Applied IQ/OQ/PQ-style qualification principles to support installation, operational, and performance evaluation of new machinery and product development processes.
- Supported improvement and quality activities by applying Ishikawa and DMAIC principles.
- Implemented mandatory Lean Six Sigma Green Belt coursework for the company, including foundational Measurement System Analysis and Gage R&R concepts.

OptoMechatroics Project, *Precision Chopper MPC Design* ***Netherlands***|*02/2025*

- Developed and analysed a precision mechatronics MPC control system for a satellite chopper, using finite-horizon prediction to achieve accurate blade positioning while rejecting vibration disturbances and compensating for hysteresis, thermal effects, and piezoelectric actuator dynamics.
- Calibrated the mechanical zero point and synchronised blade motion with internal blackbody and deep-space reference loads, using telemetry feedback to compensate for thermal and electronic drift and maintain repeatable measurement performance.

Differ Graduation Internship, *Advanced Divertor Control of Tokamak* **Netherlands** |*09/2025-02/2026*

- Developed a MATLAB toolbox for nonlinear system identification and model-based analysis of fusion-reactor control systems.
- Converted linear system representations into Lur'e-type output-feedback form to support nonlinear analysis and time-delay compensation.

- Applied to the circle criterion to analyse static nonlinearities and determine absolute-stability conditions.
- Developed transfer-function models, stability plots, and circle-criterion visualisations for control-system analysis.
- Validated analytical stability conditions in MATLAB/Simulink under nonlinear and time-delay conditions.
- Developed and documented instrumentation loop and automation loop representations for relevant control-system components.
- Supported the modeling of signals, feedback paths, sensors, actuators, and control interfaces within the system architecture.
- Delivered MATLAB/Simulink code, toolbox functions, simulations, results, and technical documentation to the DIFFER GitLab repository.
- Prepared documentation to support knowledge transfer, reproducibility, future maintenance, and reuse by subsequent project members.

Funnel Control Implementation, Project **Netherlands** | *04/2025- 06/2025*

- Implemented a funnel-control design for a second-relative-degree nonlinear system using MATLAB and Simulink. Applied the controller to a self-balancing bike model and analysed constrained tracking performance.
- Developed simulations to evaluate transient response, reference tracking, robustness, and funnel-boundary compliance. Documented the modeling, controller design, simulation setup, and results. **Funnel control video**

Academic Systems Engineering, Project **Netherlands** | *04/2025- 06/2025*

- Developed a complete systems-engineering concept in GENESYS for airline operational continuity during major disruptions, including war-related events and volcanic-eruption scenarios affecting Icelandic airspace.
- Developed requirements-engineering workflows to improve traceability between stakeholder needs, system functions, subsystem requirements, verification methods, and validation activities.
- Applied to the V-model to structure stakeholder needs, system requirements, functional decomposition, system architecture, implementation, integration, verification, and validation.
- Defined system-level scenarios, operational constraints, interfaces, risks, and response procedures for emergency airline operations.
- Prepared systems-engineering documentation, requirements, architecture descriptions, verification plans, and validation material using Microsoft Word and Microsoft Office.
- Applied with a model-based and lifecycle-oriented approach to support stakeholder coordination, system integration, and evidence-based acceptance decisions.

Applied Mathematics Program Committee, Member ***Netherlands*** | 01/10/2024 – Current

- Represented Systems and Control students as the sole student representative on a committee of 12 professors and students. Reviewed course feedback, programme structure, and student-learning outcomes.
- Contributed to improvements in model-reduction and control courses.
- Supported scheduling improvements for robust-control education in the 2026 academic year.
- Coordinated feedback between students, academic staff, and programme leadership.

LEADERSHIP & VOLUNTARY EXPERIENCE

ISVDeen, Board Member ***Netherlands*** | 01/11/2024 – 01/11/2025

- I led communications for ISVDeen, managing communication of 6 committees and 2 social chats.
- By facilitating discussions to resolve leadership tensions, helped restructure roles, contributed to record attendance of 140 at our Ramadan Iftar event.

WHO Volunteer ***Lahore, Pakistan*** | *Jun-Jul 2017*

- Successfully generated awareness drives on snow leopards' vultures, on the brink of extinction, raising *50000* Rupees by selling WHO shirts.

Waseela, *Volunteer teacher* ***Lahore, Pakistan*** | *May 2021-Sep 2022*

- I coordinated and managed a team of *6* teachers to effectively tutor underprivileged children by building valuable skills in Excel, Coding, Photoshop, and Mathematics. Goals were achieved, and the *9* students were able to acquire Copy writing, Data Entry, and Customer Service jobs.

SKILLS

Control Engineering: Control; Nonlinear Control; Robust Control; Model Based Control; Port-Hamiltonian Systems; Funnel Control; System Identification; Stability Analysis; PLC Fundamentals, Transfer-Function Modelling

Systems Engineering: V-Modelling; Systems Validation, Systems Verfication; implementation Genesys

Robotics ROS2; Python; URDF Modelling; Forward/Inverse Kinematics; Geometric Jacobians; Denavit–Hartenberg Modelling; End-Effector and 6-DOF Finger Design, VLA, MuJoCo Orchestration Sim to Real

Simulation Software: Isaac Sim/Isaac Lab; Rivz ROS2; MuJoCo

Cloud Technologies: Git/GitHub; Microsoft Office

Mathematics Software: Wolfram Mathematica MATLAB, Simulink

3D-Design Skills: ANSYS, AutoCAD; PTC Creo

Industrial Skills: Safety Management, Material Handling & Packaging, Risk Management

Business Skills: Market Research, and Stakeholder Coordination,Pattern Recognition, Business Development, Product Development

Supply Chain Management: Procurement & Sourcing,Inventory & Planning, Logistics

Testing Verification and Validation: System & Subsystem Verification, Test method development, Test management, MSA (Gage R&R), IQ/OQ/PQ, Root cause analysis, Lean Six Sigma Green Belt.

Process Engineering: Instrumentation loops, Automation Loops

Soft Skills: Market listening, stakeholder coordination, opportunity analysis, negotiation, commercial judgement, initiative, communication. Safety consciousness, supplier coordination, process thinking, quality mindset, commissioning discipline, decision-making, stakeholder management. Mathematical reasoning, model-based thinking, persistence, technical communication, learning agility, and experimental discipline. Attention to detail,

evidence-based reasoning, test discipline, impartial technical judgement, root-cause reasoning, quality mindset. Requirements judgement, stakeholder management, negotiation, structured analysis, traceability discipline, risk awareness, document Analytical problem solving, technical curiosity, debugging discipline, cross-functional collaboration, adaptability, technical communication, ownership. Notation. Systems thinking, interface communication, troubleshooting, prioritisation, practical judgement, teamwork, and resilience.

Strategic and Critical thinking, Team building, Conflict resolution

CERTIFICATIONS

Mastering ROS 2 for Robotics Programming Specialization	Coursera 2026
Modeling and Simulation with Simulink	Coursera 2026
Git, GitHub and Bash for Beginners	Physical Course at DIFFER 2025
Supplier Management for Running Business	Everest BioChem 2024
Strategic Time Management	Time Lenders 2024
Strategic Visioning	Time Lenders 2024
Six Sigma Green Belt course	LinkedIn Learning 2024
Root Cause Analysis Cause and Effect in Business	LinkedIn Learning 2024
Supply Chain Foundations	LinkedIn Learning 2024

Languages: Urdu (Mother Tongue): English (Fluent); Classical Arabic (B1); Dutch (Basic, CEFR A1)